4190.428 Introduction to Machine Learning

Support Vector Machines
(Part 1)

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Overview

- SVMs are among the best "**off-the-shelf**" **supervised** learning algorithms

- To tell the SVM story we need to know
  - **Margins** and the idea of separating data with a large "**gap**"
  - The optimal margin classifier, **Lagrange duality**
  - **Kernels**, which give a way to apply SVMs efficiently in very high dimensional (such as infinite-dimensional) feature spaces
  - **SMO algorithms**, which gives an efficient implementation of SVMs

Margins: intuition

- Consider logistic regression, where the probability $p(y = 1 \mid x; \theta)$ is modeled by

$$h_\theta(x) = g(\theta^\top x)$$

- Predict "1" on an input x if and only if $h_\theta(x) \geq 0.5$, or equivalently, if and only if

$$\theta^\top x \geq 0$$

Margins: intuition

- Consider a positive training example ($y = 1$).
  - The larger $\theta^\top x$ is, the larger also is $h_\theta(x) = p(y = 1 \mid x; \theta)$ and thus also the higher our degree of "confidence" that the label is 1
  - We can think or our prediction as being very confident that $y = 1$ if $\theta^\top x \gg 0$
- Similarly, predicting $y = 0$ if $\theta^\top x \ll 0$

Margins: intuition

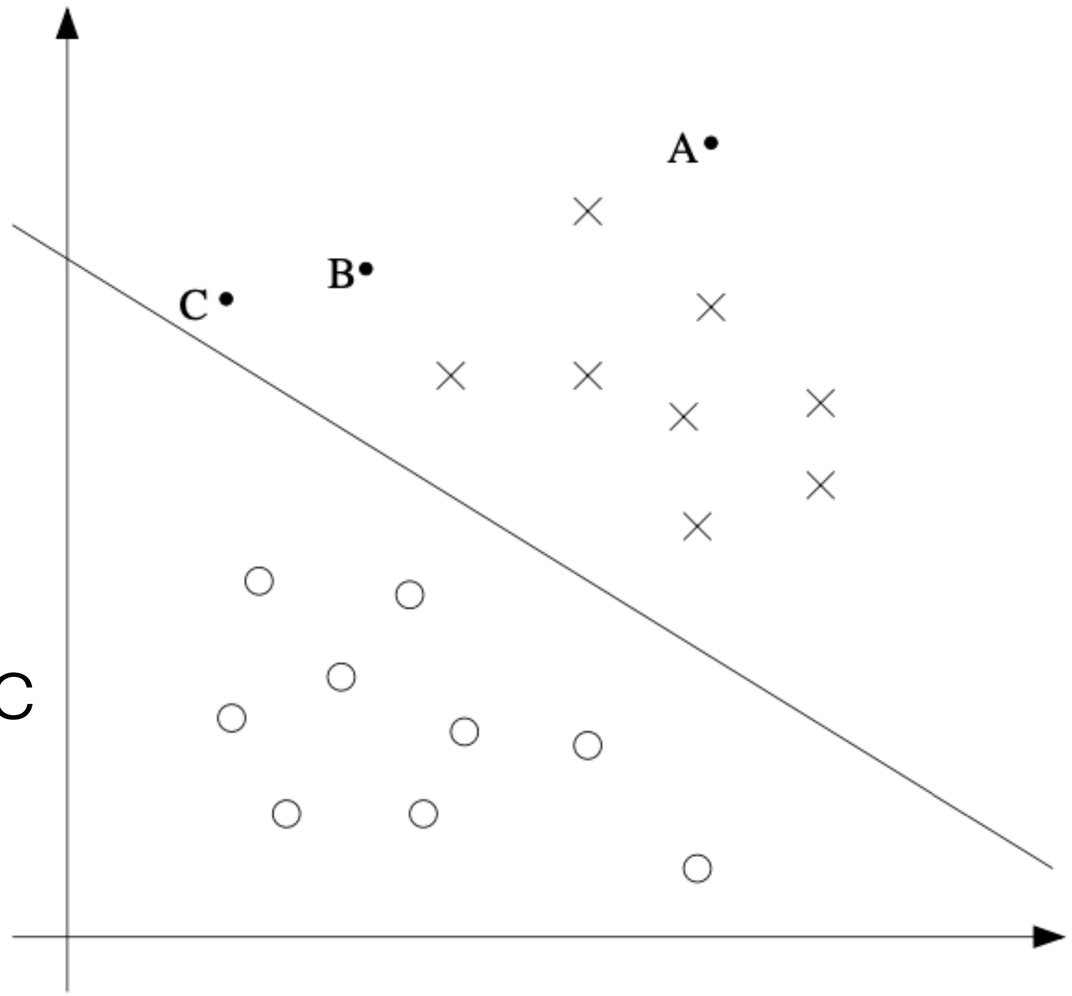
- **Good fit to the training data** :

$$\theta^\top x^{(i)} \gg 0 \text{ whenever } y^{(i)} = 1$$
$$\theta^\top x^{(i)} \ll 0 \text{ whenever } y^{(i)} = 0$$

- To be nice goal to aim for, we formalize this idea using the notion of **functional margins.**

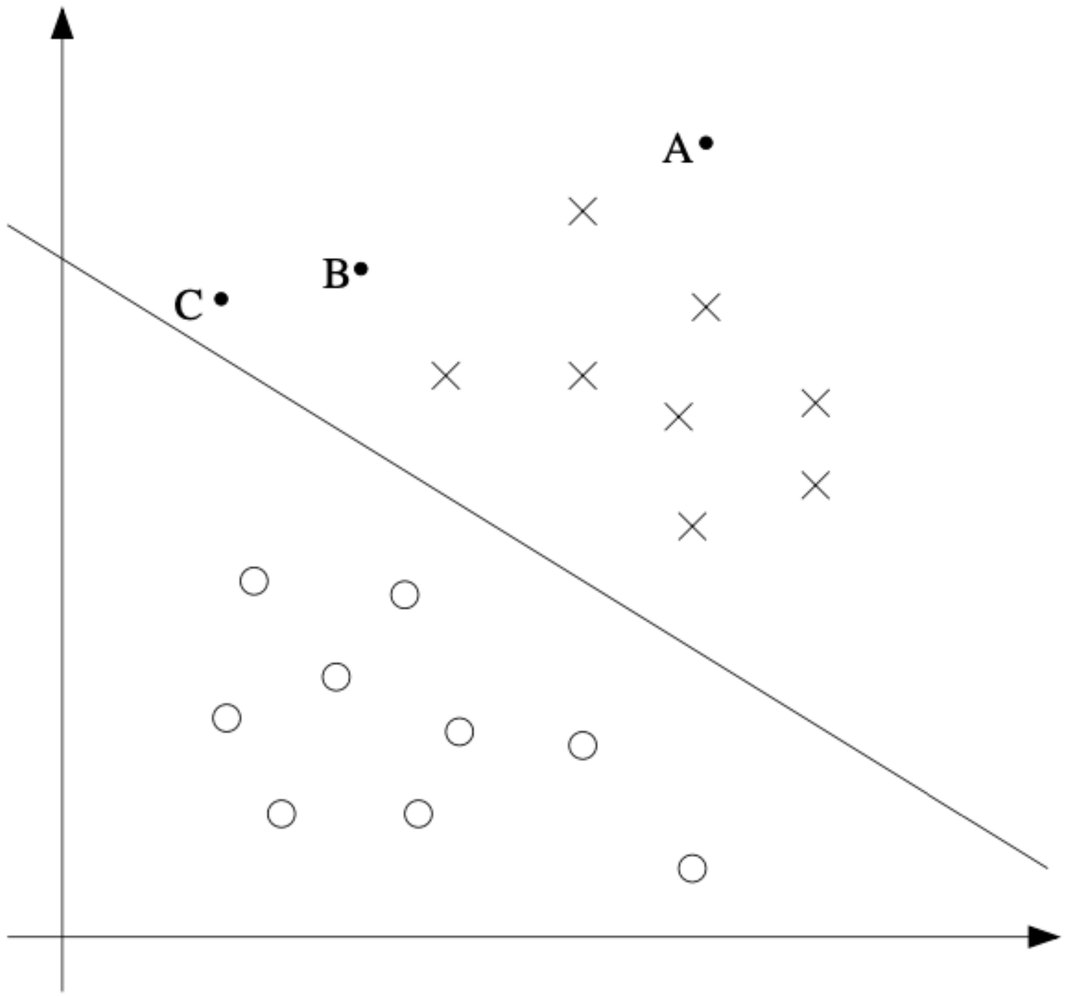
Margins: intuition

- X denote **positive** training example
- O denote **negative** training example
- Decision boundary :
  - $\theta^{\top} x = 0$ is called **separating hyperplane**
- Confident about prediction $y = 1$: A > B > C



Margins: intuition

- We manage to find a decision boundary that allows us to make all correct and confident predictions on the traing examples
- We'll formalize using the notion of **geometric margins**



Notation

- New notation for taking about classification (labels y and features x)
  - Use $y \in \{-1,1\}$ (instead of $\{0,1\}$) to denote the class labels.
  - Use parameters w, b (instead of vector θ) and write our classifier as

$$h_{w,b}(x) = g(w^\top x + b)$$

  - Definition of $g(z) = \begin{cases} 1 & \text{if } z \geq 0, \\ -1 & \text{otherwise.} \end{cases}$
  - Can explicitly treat the intercept term b separately from the other parameters
  - Thus, b takes the role of what was previously θ_0, and w takes the role of $[\theta_1 \ldots \theta_d]^\top$
- Our classifier will directly predict either 1 or -1 (cf. the perceptron algorithm), without first going through the intermediate step of estimating $p(y = 1)$ (cf. logistic regression).

Functional and Geometric Margins

- **Functional margin**:

$$\hat{\gamma}^{(i)} = y^{(i)}(w^\top x^{(i)} + b)$$

  - **Functional margin to be large** means our prediction to be confident and correct
  - If $y^{(i)} = 1$, then functional margin to be large when $w^\top x^{(i)} + b$ to be large positive numbers
  - And $y^{(i)} = -1$, functional margin to be large when $w^\top x^{(i)} + b$ to be large negative numbers
  - So, $y^{(i)}(w^\top x^{(i)} + b) > 0$ represents a correct predict

Functional and Geometric Margins

- We need **normalization**, $\|w\|_2 = 1$
  - Think about the case we replace (w, b) with $(2w, 2b)$
    - It does not affect $h_{w,b}(x) = g(w^\top x + b)$ since it only consider the 'sign'
    - However, it results in multiplying functional margin.
- It make sense to impose normalization condition (w, b) with:

$$\left(\frac{w}{\|w\|_2}, \frac{b}{\|w\|_2} \right)$$

Functional and Geometric Margins

- **Modified Functional margins**:

$$\hat{\gamma} = \min_{i=1,\ldots,n} \hat{\gamma}^{(i)}$$

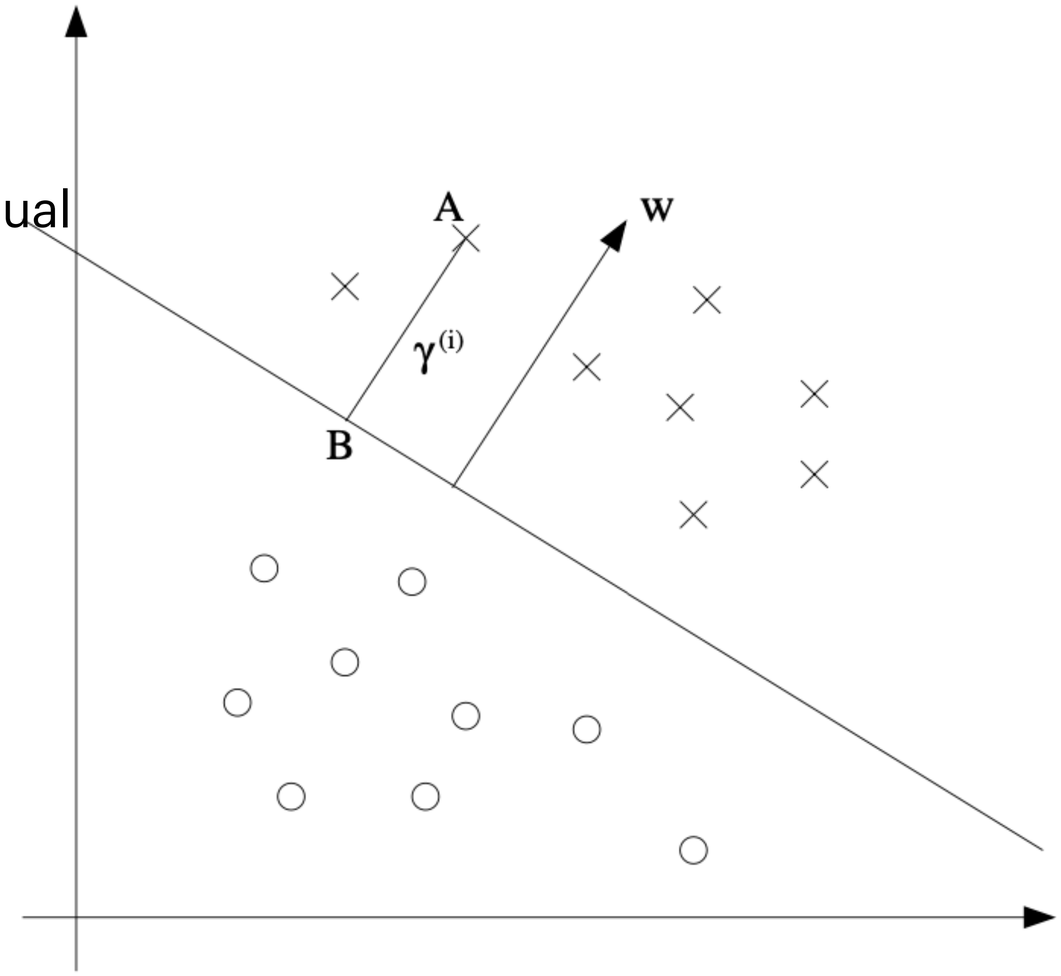
  - The smallest of the functional margins of the individual training examples

- Let A represents $x^{(i)}$, therefore B is given by (geometric meaning)

$$x^{(i)} - \gamma^{(i)} \cdot \frac{w}{\|w\|}$$

- Since B is in the decision boundary, we have

$$w^\top \left(x^{(i)} - \gamma^{(i)} \frac{w}{\|w\|} \right) + b = 0, \quad \text{[Eq.1]}$$



Functional and Geometric Margins

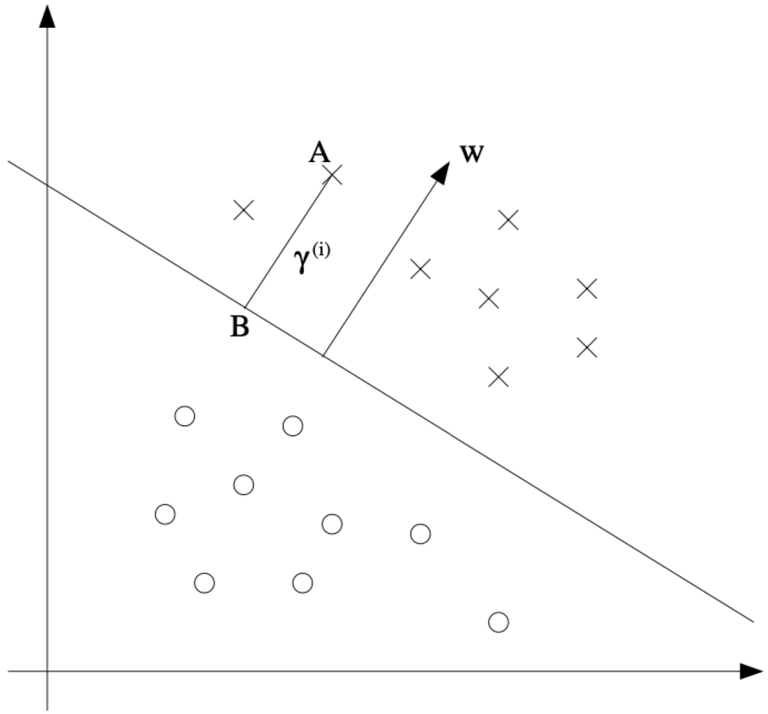
- Solving [Eq.1] for $\gamma^{(i)}$ yields:

$$\gamma^{(i)} = \frac{w^\top x^{(i)} + b}{\|w\|} = \left(\frac{w}{\|w\|}\right)^\top x^{(i)} + \frac{b}{\|w\|}$$

- We define the **geometric margin** of (w, b) with respect to a training example $(x^{(i)}, y^{(i)})$ **:**

$$\gamma^{(i)} = y^{(i)}\left(\left(\frac{w}{\|w\|}\right)^\top x^{(i)} + \frac{b}{\|w\|}\right)$$

- If $\|w\| = 1$, the functional margin equals the geometric margin
- Scale Invariance : If we rescale w and b, the geometric margin does not change.



The Optimal Margin Classifier

- The problem of maximize γ :

$$\begin{aligned} \max_{\gamma,w,b} \quad & \gamma \\ \text{s.t.} \quad & y^{(i)}(w^\top x^{(i)} + b) \geq \gamma, \quad i = 1,\ldots,n, \\ & \|w\| = 1 \end{aligned}$$

- $\|w\| = 1$ constraint is non-convex, so solving is difficult.
- We need to transform the problem

The Optimal Margin Classifier

- The problem of maximize γ:

$$\max_{\hat{\gamma},w,b} \quad \frac{\hat{\gamma}}{\|w\|}$$
$$\text{s.t.} \quad y^{(i)}(w^\top x^{(i)} + b) \geq \hat{\gamma}, \quad i = 1,\ldots,n$$

- $\|w\| = 1$ constraint is removed.
- Transform optimization problem, using $\frac{\hat{\gamma}}{\|w\|} = \frac{1}{\|w\|}$

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{s.t.} \quad y^{(i)}(w^\top x^{(i)} + b) \geq 1, \quad i = 1,\ldots,n$$

- Its solution gives us the **optimal margin classifier**
- This optimization problem can be solved using commercial **quadratic programming** (QP) code.

QnA

- Any questions?
- Note: this lecture notes are largely adopted from
  - MIT 6.036(2019 Fall), Stanford CS229(2023-2024 Summer)